OR4023 (Optimization) Homework 2

Name:_____ Student ID:_____

Question 1. Consider the problem of finding the point on the line $x_1 + x_2 = 2$ that is closest to the point $(2, 2)^T$.

a) Write the problem as a constrained optimization problem.

b) Convert the problem into an unconstrained optimization problem by eliminating x_2 from the objective function.

c) Solve the unconstrained optimization problem.

Question 2. A cylindrical can is to be designed to hold a volume of $V = 500\,\mathrm{cm}^3$. The goal is to minimize the cost of material used to construct the can, which is proportional to its surface area. The radius r must be at least 2 cm and at most 10 cm . The height h must be at least 5 cm and at most 20 cm . Formulate this problem as a constrained nonlinear optimization problem.

Question 3. Suppose f is an unimodal function in $[a, b]$ with the minimum point x^*. Let l, r be such that $a < l < r < b$. Show

a) If $f(l) > f(r)$, then $x^* \in (l, b)$.

b) If $f(l) = f(r)$, then $x^* \in (l, r)$.

Question 4. Let $f(x) = x^2 + e^{-x}, x \in [0, 1]$.

a) Show that f is unimodal.

b) Apply the exhaustive search method.

i. Determine the required number of points so that the length of the final interval of uncertainty is bounded by 0.1.

ii. Apply the exhaustive search method to find the final interval of uncertainty.

c) Apply the golden section search method.

i. Determine the required number of iterations and the number of function evaluations so that the length of the final interval of uncertainty is bounded by 0.1.

ii. Apply the golden section search method to find the final interval of uncertainty.

d) Apply the bisection method.

i. Determine the required number of iterations and the number of function evaluations so that the length of the final interval of uncertainty is bounded by 0.1.

ii. Apply the bisection method to find the final interval of uncertainty.

iii. Write a computer program in MATLAB for the bisection method and verify your answer in part (ii).